Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 2.1 · Complete a square

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in the form $(x-4)^2 + k$, with k a real number to be determined.
2. Deduce a factorisation of $f(x)$ using a special product.
3. Draw the sign table of f, then solve in $\mathbb{R}$ the inequality $f(x) \leq 0$.

Exercise 2.2 · Discussing without heavy calculation

We study the functions $f(x) = x^2 + 4$, $g(x) = (x+2)^2$ & $h(x) = -3(x-1)(x+4)$, all defined on $\mathbb{R}$.

1. For each function, determine any real roots by working directly from the form given.
2. State the sign of each function on $\mathbb{R}$. State explicitly the values at which it vanishes.
3. Explain why computing a discriminant is unnecessary in all three cases.

Exercise 2.3 · Selecting a Strategy

We want to solve in $\mathbb{R}$ the following three equations:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: special product, finding two numbers with known sum and product, or discriminant. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Check the solutions of the first two equations by substitution. For the third, check the sum and product of the roots obtained.

Exercise 2.4 · Find a parabola

The curve of a polynomial function f of degree 2 cuts the x-axis at the points with x-coordinates -2 and 3. It also passes through the point $A(1; 12)$.

1. Determine an expression for $f(x)$ from these three pieces of information. State the chosen form and why it is suitable.
2. Expand the expression found. Deduce the y-intercept of the curve.
3. Check that the function obtained satisfies the three initial pieces of information.

Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of fencing to enclose three sides of a rectangular vegetable garden. The fourth side runs along a wall and needs no fencing. Let x the length, in metres, of each of the two sides perpendicular to the wall. The whole length of fencing is used.

1. Draw a diagram and express the length of the side parallel to the wall in terms of x. State the values of x for which the rectangle is not degenerate.
2. Express the area $A(x)$, in square metres. Determine its maximum value and the corresponding dimensions, using the method of your choice.
3. Check that these dimensions use exactly 20 m of fencing.

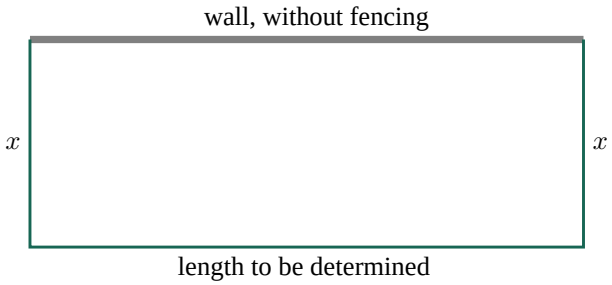


Diagram of the vegetable garden: only the three green lines use the 20 m of fencing. The proportions of the drawing do not determine the optimal dimensions.

Exercise 2.6 · A Parameter

For every real number m, consider the equation with real unknown x :

$$x^2 - 2x + m = 0.$$

The number m is a parameter: it remains fixed when the equation is solved.

1. Determine the values of m for which the equation has two distinct real solutions.
2. Determine the values of m for which it has a unique real solution, then those for which it has none.
3. Give the solution when it is unique. Explain why the three cases considered cover all values of m.

Exercise 2.7 · Link with sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for every natural number n. We seek all indices at which the term of the sequence is strictly negative.

1. First solve in $\mathbb{R}$ the inequality $x^2 - 5x + 4 < 0$.
2. Deduce all the natural numbers n such that $u_n < 0$. Justify why a real interval is not the final answer here.

Exercise 2.8 · Delayed review

A student studies the function f defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She writes: "The minimum is -5 and it is attained at $x = 3$. »

1. Say what is correct and what is wrong in this sentence. Rewrite it correctly, justifying your answer.
2. Solve in $\mathbb{R}$ the inequality $f(x) \leq -3$, then check the two bounds of the obtained set.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 43 p. 99, questions a and c. First find an obvious root, then the other using the product of the roots.

CQFD, exercise 49 p. 99, questions a and b. Express the sum and the product as an equation ; do not assume in advance that the numbers exist.